

Wipro - 100 Technical MCQ Questions

Wipro Limited

For BTech Computer Science / IT Students

Object-Oriented Programming

1. What is a 'friend' function in C++?

- A) A function that has access to private/protected members of a class
- B) A public function only
- C) A function that cannot access class members
- D) A default constructor

2. Which design principle states 'favor composition over inheritance'?

- A) SOLID-related OOP design guideline
- B) A rule that disallows interfaces
- C) A Java-only rule
- D) A database design rule

3. Which of the following is true regarding static methods and polymorphism?

- A) Static methods support runtime polymorphism
- B) Static methods are resolved at compile time and do not support runtime polymorphism
- C) Static methods cannot be called
- D) Static methods must be abstract

4. Which of these is an example of compile-time polymorphism?

- A) Method overriding
- B) Operator overloading (in C++)
- C) Dynamic method dispatch
- D) Runtime type identification

5. Which of the following correctly describes 'late binding'?

- A) Method call resolved at compile time
- B) Method call resolved at runtime based on object type
- C) A binding used only for static variables
- D) A binding used for interfaces only

6. Which OOP principle is primarily used to reduce code duplication by reusing existing classes?

- A) Inheritance
- B) Polymorphism
- C) Abstraction
- D) Encapsulation

7. In Python, which method is automatically called when an object is created?

- A) `__new__` then `__init__`
- B) `__del__`
- C) `__call__`
- D) `__str__`

8. Which of the following best describes 'abstraction'?

- A) Showing only relevant details and hiding implementation complexity
- B) Combining data and methods
- C) Creating multiple objects
- D) Executing code faster

9. In Java, which keyword is used to inherit a class?

- A) implements
- B) extends
- C) inherits
- D) super

10. Which of the following is NOT a pillar of Object-Oriented Programming?

- A) Encapsulation
- B) Polymorphism
- C) Compilation
- D) Inheritance

11. Which of these is true about interfaces in Java 8+?

- A) They can have default and static methods with bodies
- B) They can have instance variables with values changed at runtime
- C) They support constructors
- D) They cannot be implemented by multiple classes

12. What is method hiding in OOP?

- A) When a static method in a subclass has the same signature as one in the superclass
- B) When a private method is called
- C) When a method throws an exception
- D) When constructors are overloaded

13. What is object composition?

- A) Building complex objects by combining simpler objects
- B) Inheriting from multiple classes
- C) Creating static objects
- D) Deleting objects

14. What is the output-behavior difference between shallow copy and deep copy of an object?

- A) Shallow copy duplicates only references to nested objects; deep copy duplicates nested objects too
- B) There is no difference
- C) Deep copy is always faster
- D) Shallow copy always throws an exception

15. What is operator overloading?

- A) Giving additional meaning to an operator for user-defined types
- B) Removing operators from a language
- C) A compilation error
- D) A runtime exception

16. Which of these is a valid example of encapsulation in practice?

- A) Making instance variables private and exposing getters/setters
- B) Making all variables public
- C) Using global variables everywhere
- D) Avoiding classes altogether

17. Can a constructor be overridden?

- A) Yes, always
- B) No, constructors cannot be overridden
- C) Only static constructors can
- D) Only in interfaces

18. A class that cannot be instantiated directly is called a(n):

- A) Abstract class
- B) Final class
- C) Static class
- D) Sealed class

19. What is the diamond problem in OOP?

- A) An ambiguity arising from multiple inheritance of the same base class
- B) A memory leak issue
- C) A sorting algorithm's worst case
- D) A type of exception

20. Which OOP feature allows a subclass to provide a specific implementation of a method already defined in its superclass?

- A) Overloading
- B) Overriding
- C) Encapsulation
- D) Abstraction

DBMS & SQL

21. What does the term 'referential integrity' ensure?

- A) That foreign key values always match a primary key value in the referenced table (or are NULL)
- B) That all columns are indexed
- C) That all queries run faster
- D) That tables cannot be dropped

22. What is a schema in a database context?

- A) The logical structure that defines tables, columns, and relationships
- B) A backup file
- C) A stored procedure
- D) A query optimizer

23. Which normal form eliminates transitive dependency?

- A) 1NF
- B) 2NF
- C) 3NF
- D) 0NF

24. Which of the following is an example of a NoSQL database?

- A) MySQL
- B) PostgreSQL
- C) MongoDB
- D) Oracle

25. Which of these correctly describes a many-to-many relationship implementation?

- A) Implemented directly with a foreign key in one table
- B) Implemented using a junction/associative table with foreign keys to both tables
- C) Not possible in relational databases
- D) Requires a trigger

26. Which SQL keyword is used to remove duplicate rows from a result set?

- A) UNIQUE
- B) DISTINCT
- C) FILTER
- D) ONLY

27. What is a self-join?

- A) A join of a table with itself
- B) A join between two different databases
- C) A join without any condition
- D) An invalid SQL operation

28. What is the purpose of an index in a database?

- A) To enforce referential integrity
- B) To speed up data retrieval operations
- C) To store backup data
- D) To encrypt table data

29. What is a trigger in SQL?

- A) A special stored procedure automatically executed in response to certain events on a table
- B) A type of index
- C) A join operation
- D) A constraint on primary keys

30. Which isolation level allows 'dirty reads'?

- A) Serializable
- B) Repeatable Read
- C) Read Committed
- D) Read Uncommitted

31. Which SQL function is used to combine strings from multiple rows into one string (varies by RDBMS)?

- A) CONCAT() only
- B) GROUP_CONCAT()/STRING_AGG() depending on RDBMS
- C) MERGE()
- D) JOIN()

32. What does 'sharding' mean in database scaling?

- A) Splitting a large database horizontally across multiple servers
- B) Encrypting data
- C) Creating a backup
- D) Normalizing tables

33. Which of the following best defines a stored procedure?

- A) A precompiled collection of SQL statements stored in the database and executed as a unit
- B) A type of foreign key
- C) A column constraint
- D) A backup mechanism

34. Which SQL statement is used to combine the result-set of two or more SELECT statements with the same number of columns?

- A) JOIN
- B) UNION
- C) MERGE
- D) GROUP BY

35. What is the purpose of the GROUP BY clause?

- A) To sort data in ascending order
- B) To arrange identical data into groups for aggregate calculations
- C) To filter rows before aggregation
- D) To join multiple tables

36. What is the difference between WHERE and HAVING clauses?

- A) WHERE filters rows before grouping, HAVING filters groups after aggregation
- B) They are identical in function
- C) HAVING is used only with SELECT *
- D) WHERE cannot use comparison operators

Operating Systems

37. What is the primary function of an operating system?

- A) To compile programs
- B) To manage hardware resources and provide services to applications
- C) To design websites
- D) To encrypt files only

38. Which of the following best defines 'mutual exclusion' as a deadlock condition?

- A) A resource can be used by multiple processes simultaneously
- B) At least one resource must be held in a non-shareable mode
- C) Processes never request resources
- D) Resources are always preemptable

39. Which of the following best describes multiprogramming?

- A) Running multiple programs on multiple CPUs simultaneously only
- B) Keeping multiple programs in memory so the CPU always has one to execute, improving utilization
- C) Running only one program at a time
- D) A type of scheduling exclusive to real-time systems

40. Which of the following is an example of external fragmentation?

- A) Free memory scattered in small blocks between allocated partitions, none large enough for a new request
- B) Wasted space within an allocated fixed-size block
- C) A page fault
- D) A deadlock

41. What is the banker's algorithm used for?

- A) Deadlock avoidance by checking safe states before allocating resources
- B) Memory compression
- C) File compression
- D) CPU scheduling

42. Which file system attribute determines the smallest unit of disk space allocation?

- A) Sector/Block size
- B) File descriptor
- C) Inode number
- D) Directory depth

43. What is the difference between a hard link and a symbolic link in a file system?

- A) A hard link points directly to the inode; a symbolic link is a separate file containing a path to the target
- B) They are identical
- C) Symbolic links cannot be broken
- D) Hard links can span across different file systems while symbolic links cannot

44. What is a monolithic kernel?

- A) A kernel where all OS services run in kernel space as a single large process
- B) A kernel with no drivers
- C) A kernel used only in embedded systems
- D) A kernel that cannot be modified

45. What is spooling?

- A) Simultaneous Peripheral Operations On-Line - buffering data for devices like printers
- B) A type of deadlock
- C) A page replacement algorithm
- D) A CPU scheduling technique

46. What is the purpose of a context switch?

- A) To save the state of a running process and load the state of another
- B) To terminate a process
- C) To allocate memory
- D) To format a disk

47. What is an inode in a Unix-like file system?

- A) A data structure storing metadata about a file (permissions, size, pointers to data blocks) but not its name
- B) The file's name only
- C) A type of directory
- D) A kernel module

48. Which of the following is true about a microkernel architecture?

- A) It moves most services to user space, keeping the kernel minimal
- B) It has no scheduler
- C) It cannot support IPC
- D) It is always slower with no exceptions

49. What is the role of the dispatcher in CPU scheduling?

- A) To select which process runs next based purely on priority
- B) To give control of the CPU to the process selected by the short-term scheduler
- C) To manage disk I/O only
- D) To handle network requests

50. What does the term 'fragmentation' refer to in memory management?

- A) Wasted memory space due to allocation and deallocation patterns (internal or external)
- B) A type of process priority
- C) A CPU scheduling delay
- D) A file system backup issue

51. Which of these is a necessary condition to prevent deadlock by breaking 'circular wait'?

- A) Impose a strict ordering of resource acquisition
- B) Allow unlimited resource requests
- C) Never release resources
- D) Increase the number of processes

52. Which scheduling algorithm gives the shortest average waiting time (in theory, for a given batch)?

- A) FCFS
- B) SJF (Shortest Job First)
- C) Round Robin with a large quantum
- D) Priority Scheduling

Computer Networks

53. Which of the following best describes a CDN (Content Delivery Network)?

- A) A distributed network of servers that delivers content to users from a location geographically closer to them
- B) A type of firewall
- C) A DNS resolution protocol
- D) A database replication technique

54. What is the purpose of a subnet?

- A) To divide a large network into smaller, more manageable segments
- B) To increase the number of MAC addresses
- C) To eliminate the need for routers
- D) To replace DNS

55. What is the primary purpose of a load balancer?

- A) To distribute incoming network traffic across multiple servers to improve reliability and performance
- B) To encrypt data at rest
- C) To act as a firewall only
- D) To assign static IP addresses

56. What does a 'ping' command test?

- A) Reachability and round-trip time to a host using ICMP echo requests
- B) DNS resolution only
- C) Bandwidth of a connection
- D) Server uptime history

57. What is a MAC address?

- A) A unique hardware identifier assigned to a network interface
- B) An IP address used only for routing
- C) A type of DNS record
- D) A session identifier

58. Which protocol is used for secure remote login to a server?

- A) Telnet
- B) SSH
- C) FTP
- D) SMTP

59. What is the function of a DNS server?

- A) To translate human-readable domain names into IP addresses
- B) To assign private IP addresses within a LAN
- C) To encrypt HTTP traffic
- D) To manage email delivery

60. Which of the following best describes latency in networking?

- A) The amount of data transferred per second
- B) The delay before data transfer begins following an instruction
- C) The number of devices on a network
- D) The physical distance between two devices

61. Which of the following is a connectionless protocol?

- A) TCP
- B) UDP
- C) FTP
- D) HTTP over TCP

62. Which of the following is true about IPv6 compared to IPv4?

- A) IPv6 uses 32-bit addresses while IPv4 uses 128-bit
- B) IPv6 uses 128-bit addresses, providing a vastly larger address space than IPv4's 32-bit addresses
- C) IPv6 is identical to IPv4 in address length
- D) IPv4 has replaced IPv6 entirely

63. Which of the following is an application-layer protocol?

- A) TCP
- B) IP
- C) HTTP
- D) Ethernet

64. What does VPN stand for and what is its main purpose?

- A) Virtual Private Network - creates a secure, encrypted connection over a public network
- B) Virtual Public Network - increases internet speed
- C) Verified Private Node - a type of router
- D) Virtual Protocol Number - an IP addressing scheme

Data Structures & Algorithms

65. What is a monotonic stack typically used for?

- A) Finding the next greater/smaller element efficiently
- B) Sorting an array in place
- C) Implementing recursion
- D) Hashing

66. What is the time complexity of the Two-Pointer technique on a sorted array to find a pair with a given sum?

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

67. What is memoization?

- A) Storing the results of expensive function calls and reusing them when the same inputs occur again
- B) A sorting technique
- C) A type of recursion without a base case
- D) A method to delete duplicate array elements

68. Which of the following is a common technique to resolve hash collisions?

- A) Chaining or open addressing
- B) Binary search
- C) Recursion
- D) Memoization

69. What is the time complexity of finding the minimum element in an unsorted array?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

70. What is the key property of a min-heap?

- A) The parent node is always greater than its children
- B) The parent node is always smaller than or equal to its children
- C) It must be a complete binary tree only, with no ordering
- D) It is always a balanced BST

71. What is a hash collision?

- A) When two different keys map to the same hash bucket/index
- B) When a hash table runs out of memory
- C) When a key is deleted incorrectly
- D) When two hash tables merge

72. What is the time complexity of Bubble Sort in the worst case?

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(n^2)$
- D) $O(\log n)$

73. Which of the following best describes a trie data structure?

- A) A tree-like structure used for efficient retrieval of strings, especially for prefix matching
- B) A type of balanced binary search tree
- C) A hashing algorithm
- D) A graph traversal method

74. Which graph algorithm is used to find the shortest path in a graph with possible negative edge weights (no negative cycles)?

- A) Dijkstra's Algorithm
- B) Bellman-Ford Algorithm
- C) Prim's Algorithm
- D) Kruskal's Algorithm

75. Which data structure underlies the implementation of recursion internally?

- A) Queue
- B) Stack (call stack)
- C) Heap (memory heap)
- D) Graph

76. What is the time complexity of Floyd-Warshall algorithm for all-pairs shortest paths?

- A) $O(V^2)$
- B) $O(V^3)$
- C) $O(E \log V)$
- D) $O(V + E)$

77. What is the amortized time complexity of appending an element to a dynamic array (like Python's list or Java's ArrayList)?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$ amortized
- D) $O(n^2)$

78. Which algorithm is used to find the Minimum Spanning Tree of a graph?

- A) Dijkstra's Algorithm
- B) Kruskal's or Prim's Algorithm
- C) Bellman-Ford Algorithm
- D) Floyd-Warshall Algorithm

79. What is the primary difference between a greedy algorithm and dynamic programming?

- A) Greedy makes a locally optimal choice at each step without reconsidering, DP considers overlapping subproblems and often guarantees a globally optimal solution
- B) They are the same technique
- C) DP never uses recursion
- D) Greedy always gives an optimal solution while DP does not

80. Which data structure would you use to check if a string of brackets is balanced?

- A) Queue
- B) Stack
- C) Array
- D) Linked List

81. Which technique explores all possible options and backtracks when a solution path fails, often used in puzzles like N-Queens?

- A) Greedy algorithm
- B) Backtracking
- C) Dynamic programming
- D) Binary search

82. What is the average time complexity of searching in a well-implemented hash table?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n^2)$

83. What is the time complexity of building a heap from an unsorted array (heapify)?

- A) $O(n \log n)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

84. Which of the following problems is a classic example solved using dynamic programming?

- A) Binary search
- B) 0/1 Knapsack problem
- C) Bubble sort
- D) Linear search

Programming Language Fundamentals

85. What does 'pass by reference' mean?

- A) A copy of the variable's value is passed to the function
- B) The function receives a reference to the actual variable, so changes affect the original
- C) Variables cannot be passed to functions
- D) Only primitive types can be passed

86. Which of the following is a value type (not a reference type) in Java?

- A) int
- B) String
- C) Array
- D) Object

87. In C, what does the 'malloc' function do?

- A) Deallocates memory
- B) Allocates a block of memory dynamically
- C) Initializes a variable to zero
- D) Declares a static variable

88. Which keyword is used to handle exceptions in Python?

- A) catch
- B) except
- C) rescue
- D) handle

89. In Java, what is autoboxing?

- A) Automatic conversion between primitive types and their corresponding wrapper classes
- B) A way to box multiple objects together
- C) A garbage collection technique
- D) A type of casting error

90. What is the default value of an uninitialized boolean instance variable in Java?

- A) true
- B) false
- C) null
- D) 0

91. What is a memory leak?

- A) When allocated memory is never freed even though it is no longer needed, wasting resources over time
- B) When a program uses too little memory
- C) A type of compilation error
- D) A CPU scheduling issue

92. In Python, which of the following is a mutable data type?

- A) Tuple
- B) String
- C) List
- D) int

93. What does 'immutable' mean for an object (e.g., a String in Java)?

- A) Its value can be changed after creation
- B) Its state cannot be changed after it is created
- C) It can only be used once
- D) It cannot be passed to functions

94. What is the difference between a shallow copy and a deep copy of a list containing nested lists?

- A) A shallow copy also copies nested objects recursively, deep copy does not
- B) A shallow copy copies references to nested objects, deep copy recursively copies nested objects too
- C) They behave identically
- D) Deep copy is not possible in most languages

Web & Cloud Fundamentals

95. What is the primary difference between HTTP and HTTPS?

- A) HTTPS uses encryption (SSL/TLS) to secure data in transit while HTTP does not
- B) HTTP is faster in all cases
- C) HTTPS cannot use port 443
- D) There is no difference

96. What is the purpose of a Content Delivery Network (CDN)?

- A) To cache and serve content from servers geographically closer to users, reducing latency
- B) To encrypt all web traffic
- C) To replace DNS servers
- D) To manage database transactions

97. What is auto-scaling in cloud computing?

- A) Automatically adjusting compute resources up or down based on demand
- B) Manually adding servers only
- C) A type of database indexing
- D) A billing model only

98. What does 'statelessness' mean in the context of REST APIs?

- A) The server stores client session state between requests
- B) Each request from a client contains all the information needed to process it, with no reliance on stored server-side session state
- C) The API has no state at all, including no data
- D) Stateless APIs cannot use authentication

99. What does HTTP stand for?

- A) HyperText Transfer Protocol
- B) HighText Transmission Protocol
- C) Hyperlink Transfer Text Protocol
- D) HyperText Trade Protocol

100. What is a RESTful API?

- A) An API that strictly requires SOAP protocol
- B) An architectural style for designing networked applications using stateless operations over standard HTTP methods
- C) A type of database
- D) A programming language

Answer Key

1-A	2-A	3-B	4-B	5-B	6-A	7-A	8-A	9-B	10-C
11-A	12-A	13-A	14-A	15-A	16-A	17-B	18-A	19-A	20-B
21-A	22-A	23-C	24-C	25-B	26-B	27-A	28-B	29-A	30-D
31-B	32-A	33-A	34-B	35-B	36-A	37-B	38-B	39-B	40-A
41-A	42-A	43-A	44-A	45-A	46-A	47-A	48-A	49-B	50-A
51-A	52-B	53-A	54-A	55-A	56-A	57-A	58-B	59-A	60-B
61-B	62-B	63-C	64-A	65-A	66-C	67-A	68-A	69-C	70-B
71-A	72-C	73-A	74-B	75-B	76-B	77-C	78-B	79-A	80-B
81-B	82-C	83-B	84-B	85-B	86-A	87-B	88-B	89-A	90-B
91-A	92-C	93-B	94-B	95-A	96-A	97-A	98-B	99-A	100-B

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.